21/3/25 Lec - 5

DIGITAL

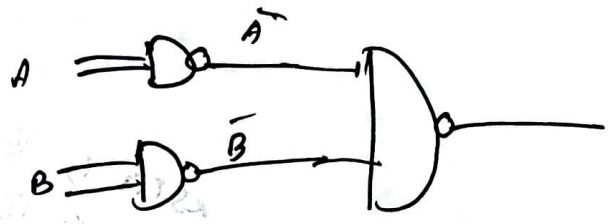
$\overline{A+B} = \bar{A} \cdot \bar{B}$

$\overline{AB} = \bar{A} + \bar{B}$

$A \oplus B = A\bar{B} + B\bar{A}$

To check 2 signals if same → XOR

⇝ OR gate from NAND



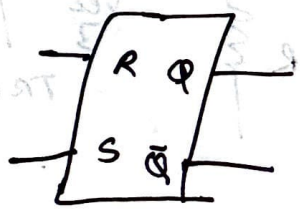
$\overline{\bar{A} \cdot \bar{B}} = \underbrace{A+B}_{\text{OR Gate}}$

Latches

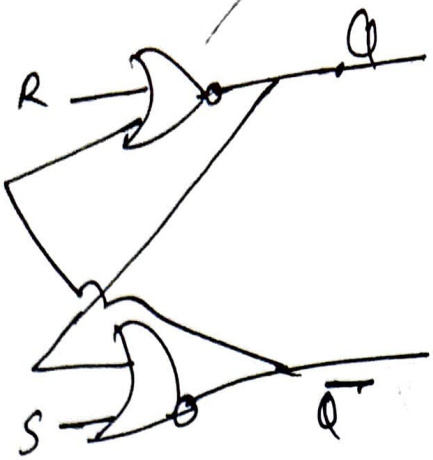
1. SR Latch ★
2. D Flip flop
3. JK flip flop

SR Latch

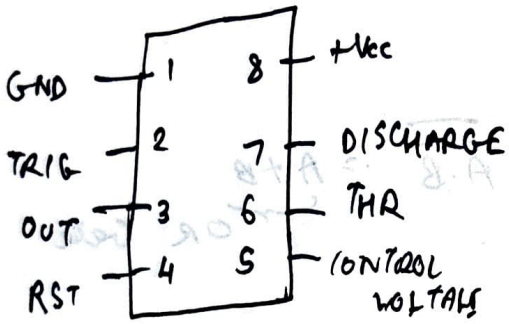
R	S	Q	$\bar{Q}$
1	0	0	1
0	1	1	0
0	0	Retain whatever is	
1	1	Not Defined	



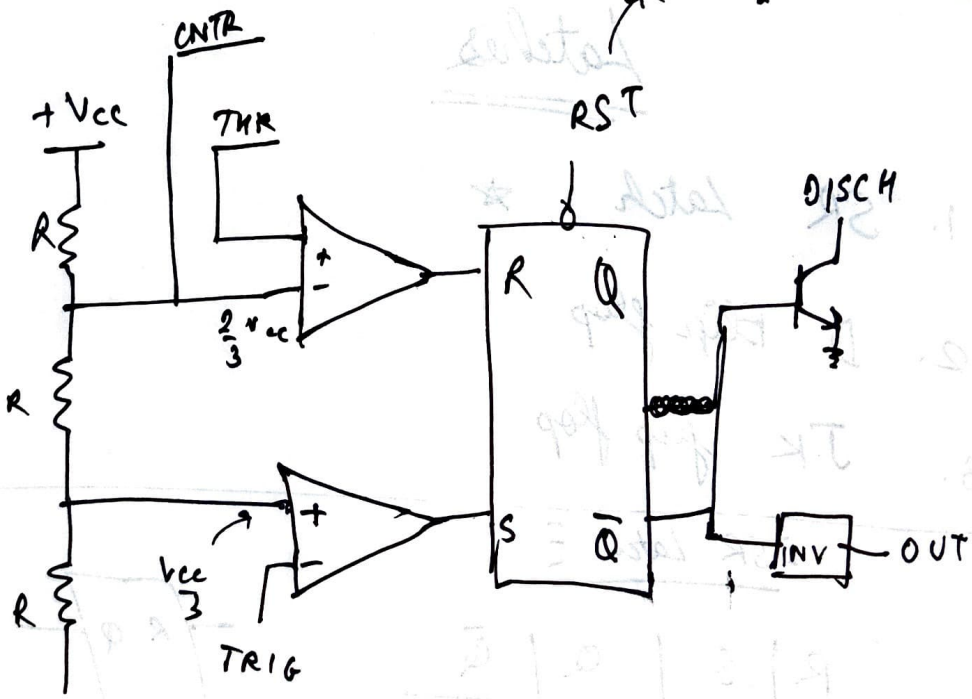
Using NOR Gates: Same with NAND Do



Timers: NE 555

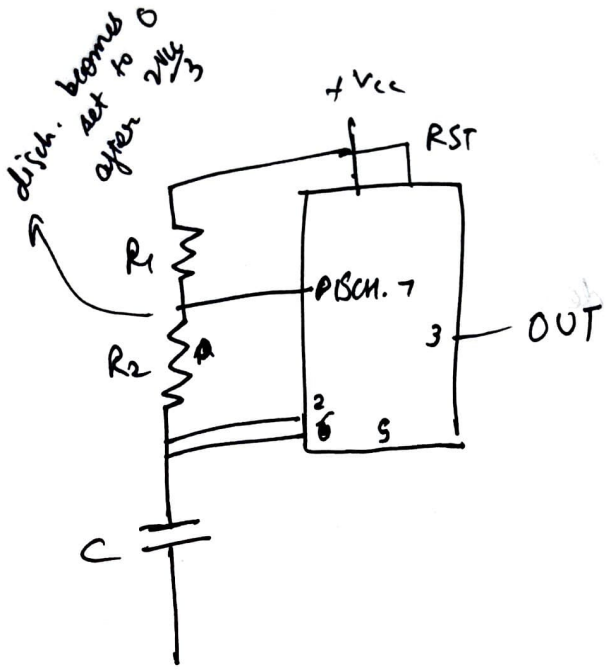


(when RST is 1, Q is 1
" is 0, Q is 0)



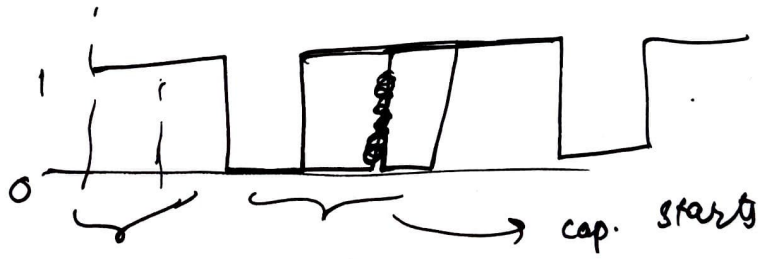
Imp.

THR	TRIG	RST	R	S	Q
$< \frac{2}{3}V_{cc}$	$< \frac{V_{cc}}{3}$	1	0	1	1
$< \frac{2}{3}V_{cc}$	$> \frac{V_{cc}}{3}$	1	0	0	Stable
$> \frac{2}{3}V_{cc}$	$> \frac{V_{cc}}{3}$	1	1	0	0
×	×	0	×	×	0



1. Astable mode

Out :



cap. starts discharging

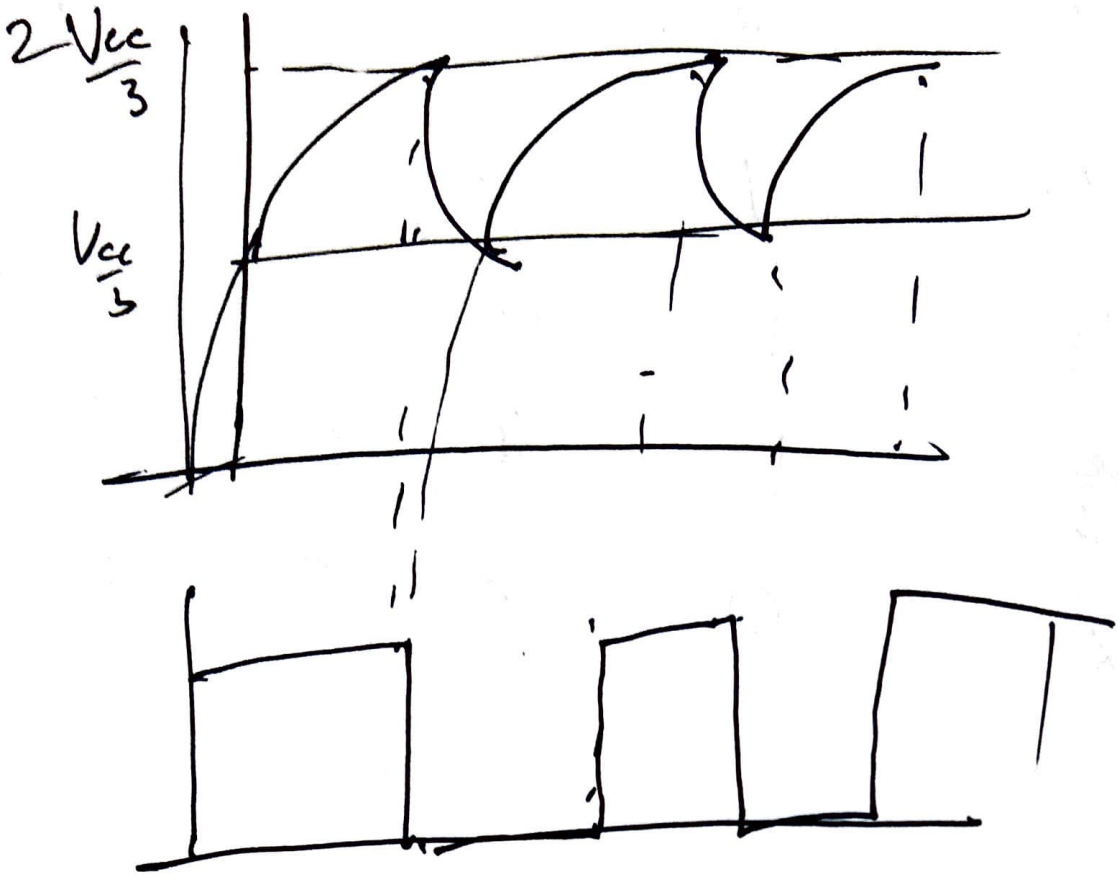
Till when $V_c = \frac{2V_{cc}}{3}$

($\frac{V_{cc}}{3}$ → sets to 1

$\frac{2V_{cc}}{3}$ → stable)

till $\frac{V_{cc}}{3}$ → sets to 0

then $\frac{2V_{cc}}{3}$ → sets to 1



2. Bistable mode

3. Monostable mode